SAYED AHMED HUSAIN

Electrical Engineer | Full-stack Developer | "Solar PV - CRPEP – Wireman" Licensed

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SUMMARY

Electrical Engineer and Sales Strategist with specialized expertise in **Electrical Engineering for Fire Safety Systems**. I bridge the gap between complex technical requirements and business growth, leveraging deep knowledge of **NFPA standards** and **fire alarm system design** to secure high-value contracts and manage government tenders across Bahrain.

Currently advancing into **Full-Stack Development**, I apply this same engineering precision to build robust, scalable software solutions using Go and JavaScript. My goal is to leverage this dual-competency—industrial engineering insight combined with modern software architecture—to develop efficient tools that solve complex real-world challenges.

TECHNICAL SKILLS

- **Core Engineering:** Fire Alarm System Design (NFPA/GCC Codes), Electrical Installation, Solar PV Systems.
- **Engineering Software:** AutoCAD, MATLAB/Simulink, Electronics Circuit Design.
- **Languages:** Go (Golang), TypeScript, JavaScript, Python, SQL (SQLite), HTML/CSS.
- **Web Technologies:** React, Flutter, Next.js, Node.js, Tailwind CSS, Apollo Client, WebSockets, GraphQL.
- **Systems & Tools:** Docker, Git, AWS, Postman, TCP/IP, JSON, REST APIs, OAuth2, JWT.
- **Computer Science Concepts:** Algorithms & Data Structures, Concurrency & Parallelism, Virtual DOM, State Management.

PERSONAL SKILLS

Adaptability, Communication, Fast Learner, Decision Making, Teamwork, Problem-solving, Time Management, Leadership, Training and Development, Project Management.

PROFESSIONAL EXPERIENCE

Public Relations In-Charge | Lichfield Fire and Safety Equipment Co., Ltd. (LIFECO)

Bahrain/UAE | May 2024 — February 2026

- **Strategic Sales Leadership:**
  - Formulate and execute comprehensive **Strategic Sales** plans and quotas for the Bahrain market, ensuring strict alignment with corporate management objectives.
  - Drive profitability by analyzing market dynamics, tracking competitor activities, and identifying new opportunities to increase revenue and market share.
  - Lead and mentor sales teams, setting clear targets and monitoring performance to ensure sales goals are consistently met or exceeded.
- **Tender & Procurement Management:**
  - Secure high-value deals with public and private sectors by actively managing bids on the **Bahrain Tender Board** and **Procural (Tenders & Auctions Platform)**.
  - Oversee the end-to-end **Vendor Management** process, obtaining and maintaining industrial vendor pre-qualifications with key government and private entities.
  - Identify and pursue public and private **Government Tenders**, managing the submission process to ensure compliance and competitiveness.
- **Client Relations & Business Growth:**
  - Cultivate robust relationships with key accounts and expand the customer base by identifying and counseling district sales representatives.
  - Conduct regular client visits to diagnose evolving needs, assess relationship quality, and negotiate pricing to close and renew orders.
  - Establish and nurture critical relationships with **Civil Defense** and government ministries to facilitate smooth project execution and compliance.
- **Market Intelligence & Operations:**
  - Monitor key economic indicators, emerging market trends, and supply and demand shifts to optimize product mix and sales volume.
  - Recommend new product lines and service improvements based on comprehensive market analysis and consumer needs surveys.
  - Maintain technical expertise by participating in professional networks and reviewing industry publications to stay ahead of the curve.

Electrical Engineer | Super Fire and Safety Services W.L.L

Bahrain | October 2022 — October 2023

- **System Design & Engineering:** Completed detailed **Fire Alarm System Designs** and technical drawings utilizing current Bahrain CDFSD, GCC Fire Codes, **NFPA standards**, and project specifications.
- **Installation & Commissioning:** Directed electrical teams in the installation, programming, maintenance, testing, and commissioning of fire alarm control panels and suppression systems, ensuring full compliance with NFPA and EWA regulations.

- **Quality Assurance:** Conducted rigorous inspections and system tests to ensure all components met **Electrical Engineering** safety requirements and performance standards.
- **Key Projects:**
  - **Bahrain Aluminium Company (ALBA):** Deluge fire fighting system.
  - **Bahrain Steel Company:** Fire suppression systems.
  - **Ministry of Education:** Fire alarm systems.
  - **MCSIX:** Fire alarm and firefighting systems for the Ministry of Works STP.

Retail Sales Specialist | Channels by STC

Bahrain | April 2022 — September 2022

- **Sales Strategy:** Served as a Retail Sales Advisor, expertly executing cross-selling and upselling strategies for a wide range of telecommunications products and services.
- **Customer Management:** Managed complex customer inquiries and resolved complaints efficiently using the Oracle CRM enterprise system, ensuring high levels of customer satisfaction and retention.

Retail Marketing and Customer Service | Zain Bahrain B.S.C

Bahrain | February 2019 — March 2022

- **Operations Management:** Served as Acting Branch Manager, overseeing sales team performance and optimizing retail shop operations to meet company standards.
- **Sales Excellence:** consistently exceeded targets by cross-selling and upselling telecommunication services and products as a frontline agent.
- **Technical Support:** acted as a Service Center Representative, troubleshooting service issues and resolving technical claims to minimize downtime for customers.
- **Customer Engagement:** Resolved high-volume customer complaints and requests with a focus on problem resolution and conducted outbound tele-sales calls to drive promotion uptake.

EDUCATION

Full Stack Development Specialized Program (High Diploma)

Reboot01 Coding Institute, Bahrain | March 2024 — March 2026

Bachelor of Science in Electrical Engineering

University of Bahrain, Bahrain | June 2022

High School (Physics and Mathematics)

H.E Sheikh Isa bin Ali Secondary School, Bahrain | May 2015

LICENSES & CERTIFICATIONS

- **NOCN Level 3 Award for Solar PV Installer and Operator** ***[Certificate]***
  - **Issuing Authority:** NOCN
  - **ID:** 48282023
- **Certificate of Acknowledgement (Solar PV)** ***[Certificate]***
  - **Issuing Authority:** Ministry of Electricity and Water Affairs (Bahrain)
  - **MEWA Certificate No:** 2024-21
  - **Issued:** Jul 2024 | **Expires:** Jul 2029
- **CRPEP License (Electrical)**
  - **Issuing Authority:** CRPEP (Bahrain)
  - **ID:** EPP/C4597/EL/01-D
  - **Issued:** Dec 2022 | **Expired:** Dec 2023
- **Electrical Installation License**
  - **Issuing Authority:** EWA (Bahrain)
  - **ID:** W2402010
  - **Issued:** Feb 2024 | **Expires:** Feb 2029
- **Basic Life Support (BLS)**
  - **Issuing Authority:** AHA
  - **ID:** 225607015422
  - **Issued:** Jul 2022 | **Expired:** Jul 2024
- **Course Certificates:**
  - B2B Sales Fundamentals
  - Emotional Intelligence
  - AI in Business Essentials
  - Leading with Innovation
  - Critical Thinking

PROJECTS PORTFOLIO

Smart Streetlights | University Graduation Project *[Thesis, Poster]*

January 2020 — May 2020

- The current streetlight control methods in Bahrain lack precision and environmental sensing, prompting engineers to reassess the standard to find cost-effective solutions for smart road lighting. The project aims to develop an intelligent lighting system that improves the existing streetlight functionality.
- To address this, an Arduino controller and motion detection sensors will be used to activate streetlights based on traffic conditions. Additionally, the project will compare energy consumption between high-pressure sodium lights and LEDs.
- Implementing an intelligent street lighting system saves energy, extends lighting lifespan, and reduces maintenance costs. It optimizes lighting by avoiding unnecessary usage when no vehicles or pedestrians are present.

GigaBit — Real-Time Social Network (Go + Next.js) *[Repository]*

- Built GigaBit, a **Full-Stack Development** project featuring a real-time social networking platform that covers the complete surface area of a modern community application, from authentication and privacy controls to live communication and group collaboration.
- **Key Features:**
  - Profiles with public/private visibility, follow requests, and approval flows.
  - Content system including posts with media, threaded comments, reactions, bookmarks, shares, and polls.
  - Community features such as groups with roles and permissions, invitations, gated membership, group posts, and events with RSVP handling.
  - Real-time capabilities using WebSockets for private and group messaging, typing indicators, notifications, online status, and live feed updates.
  - Relational data model using SQLite to support the social graph and interaction workflows.

01 Dashboard – Personal Learning Analytics Platform (React + Typescript + GraphQL) *[Repository]*

- 01 Dashboard is a web-based analytics dashboard built to visualize and analyze learning progress on the reboot01 platform. The project aggregates data related to XP, projects, audits, and timelines, presenting it in a clear and structured way to help users better understand their learning patterns over time.
- The dashboard consumes real data through a GraphQL API and uses Apollo Client for efficient querying, caching, and request batching. Authentication is secured using JWT, with token refresh logic to maintain stable user sessions.
- From a front-end perspective, the project focuses on clarity and usability. Interactive charts and visual summaries are used to represent progress, audit performance, and project distribution, while a responsive layout ensures the dashboard works well across devices.

Connecthub RT – Real-Time Forum & Chat Application *[Repository]*

- Connecthub RT is a real-time web application built with Go that combines a traditional discussion forum with live communication features. The platform supports posts, comments, categories, and private messaging, allowing users to interact without the delays typically found in standard forum systems.
- The backend is structured around Go's concurrency model and WebSocket communication, enabling instant updates for comments, online status, and private chats. Persistent data such as users, posts, and messages is managed using SQLite, keeping the architecture simple while still reflecting real-world application design.
- The project focuses on practical backend concepts, including session handling, secure password hashing, and real-time connection management. The frontend uses HTML, CSS, and JavaScript to deliver a responsive interface that works smoothly across devices.

PicoJS Framework — Reactive UI Framework (JavaScript) *[Repository]*

- Built PicoJS Framework, a lightweight JavaScript framework created to explore and implement the core mechanics behind modern UI libraries in a clean and readable codebase. The project focuses on understanding how reactive state, Virtual DOM rendering, and event handling work together to keep user interfaces predictable and efficient.
- **Key Highlights:**
  - Virtual DOM diffing with minimal DOM patching.
  - Reactive state management with automatic re-renders.
  - Centralized event delegation for better performance and cleanup.
  - Hash-based routing that stays in sync with application state.
  - Includes a complete TodoMVC demo demonstrating real-world usage.

01 ConnectHub – Full-Stack Web Forum Application *[Repository]*

- 01 ConnectHub is a full-stack web forum application built with Go, tailored to support structured community discussions and meaningful user interaction. The platform allows users to register, create posts, participate in discussions through comments, and engage with content using likes and dislikes, all within a moderated and secure environment.
- The application implements a complete authentication system, including secure password hashing and OAuth2 social login using Google and GitHub. Role-based access control enables administrators and moderators to manage users and content effectively.
- From a technical perspective, the project focuses heavily on backend architecture, session management, and data persistence. Server-side rendering is powered by Go, while the frontend is built with HTML, CSS, and JavaScript.

HarmonyHub – Music & Concert Discovery Platform *[Repository]*

- HarmonyHub is a web application engineered in Go that brings together music artists and their concert information in a single, structured platform. The application aggregates artist profiles, band members, formation details, and performance schedules using data retrieved

from an external REST API.
- The project focuses on clean server-side architecture, structured data handling, and clear presentation of API-driven content. Artist and concert data are processed using Go structs and rendered through server-side templates, ensuring reliability, performance, and maintainability.

AntFinder – Graph Pathfinding Simulation in Go *[Repository]*

- AntFinder is a terminal-based application programmed in Go that simulates efficient pathfinding for ant colonies moving through a graph. The project focuses on discovering optimal routes and coordinating the movement of multiple ants from a start node to an end node while minimizing the total number of turns.
- The application applies classic graph algorithms, including Depth-First Search (DFS), to explore and evaluate possible paths. Ant movement is simulated step by step, ensuring conflict-free traversal and balanced path assignment.

GoTalk – Concurrent TCP Chat Server *[Repository]*

- GoTalk is a TCP-based chat server architected in Go, optimized to support real-time communication between multiple terminal clients. The application allows users to connect concurrently, exchange messages instantly, and interact within a shared chat environment.
- The project focuses on core systems programming concepts, including concurrent client handling using goroutines, message broadcasting through shared channels, and reliable network communication via TCP sockets. Each message is timestamped, users are uniquely identified, and chat activity is persisted through server-side logging.

Space Blaster – Browser-Based Arcade Game *[Repository]*

- Space Blaster is a browser-based space shooter built using HTML, CSS, and JavaScript, inspired by classic arcade games. Players control a spaceship to navigate enemy waves, fire lasers, and build their score in real time.
- The project emphasizes smooth gameplay, responsive keyboard controls, and clear game-state management. Enemy movement, collision detection, scoring, and lives are orchestrated through structured JavaScript logic and DOM manipulation. Audio effects are integrated to enhance immersion without relying on external libraries.

ASCII Forge – ASCII Art Generator Web Application *[Repository]*

- ASCII Forge is a web-based application implemented in Go that converts plain text into stylized ASCII art using multiple banner formats, including standard, shadow, and thinkertoy. The application provides a clean and intuitive interface that allows users to generate creative text representations directly from the browser.
- The project focuses on server-side text processing and dynamic content generation, handling banner parsing and rendering logic efficiently. Users can export the generated ASCII output as text files, making it easy to reuse or share across different platforms.